MINCHAN KIM

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EDUCATION

Qingdao Daewon School, Qingdao, China, *Class of 2027* Aug 2024 to June 2027 (expected)

- Unweighted GPA 3.984 / 4.0. Intended study: artificial intelligence and related fields.
- AP exams planned for May 2027: Calculus BC, Physics C (Mechanics; Electricity and Magnetism), Physics 1, Computer Science A.

HONORS AND AWARDS

All awards below were issued by Qingdao Daewon School and confirmed in the school's record of awards.

- Research and science: 1st Place, Excellent Research Paper, Science and Technology (Dec 2025); 1st Place, Science Fair, High School Division (Apr 2026).
- Subject Excellence Award: Precalculus (Dec 2025, Jul 2026); Creative Writing (Dec 2025).
- Academic Improvement Award: Economics (Dec 2024), English Academic Reading (Dec 2024, Jul 2026), Economics 10 and English Language and Composition (Jul 2025), Physics (Dec 2025), Chinese 11 (Jul 2026).
- Contests: 1st Place, English Speaking Contest (Nov 2024); 2nd Place, school-wide Korean Literature Writing Contest, prose (May 2025); 1st Place, Social Studies Project Creativity (Jun 2025); Popularity Award, Project Creativity, Chinese (Oct 2025).
- Chinese Month: 2nd Place, Poster Making (Sep 2024); 3rd Place, Paper-Weaving and Chinese Calligraphy Contest (Sep 2025); 3rd Place, Poster Making (Mar 2026).
- Performance Excellence Award: Point of View club (Jul 2025); COSMOS (Jul 2026). Certificate of Participation, DAEMUN II (Sep 2025).
- Sports (1st Place): men's basketball and futsal (May 2025); kickball (Jun 2026).

RESEARCH AND PROJECTS

F-22-Inspired RC Aircraft: Design and Flight Stability Sep to Dec 2025

- Designed and built a corrugated-cardboard RC aircraft from hand-drawn patterns, flight-tested it, and wrote a 22-page paper relating design variables to flight stability. Won 1st Place in the school research paper competition.

Mechanism of Autopilot (school physics paper)

- Analyzed a control-systems paper and an Airbus patent on autopilot disengagement alerts; drew the autopilot-to-manual transition logic and compared illustrative P, PI and PID error curves. Literature-based, written with teacher guidance; no flight or hardware testing.

Environmental ESG Initiatives at Embry-Riddle University (school environmental science paper)

- Wrote a short case study from the university's public materials on flight training, sustainable aviation fuel, campus facilities and research funding. No original data.

Tern, flight-search web app (flytern.site) Apr 2026 to present

- Built with Next.js, the Duffel API and React Three Fiber in English, Korean, Japanese and Chinese; compares fares, duration, layovers and airport weather, with a rule-based schedule-risk score. Checkout runs through an external partner; no revenue.

Automated Drone-Tracking Turret (school science fair)

- Built a camera-tracking rig with Python, OpenCV and Arduino-driven servos; tested YOLOv5 detection and a color-tracking dual-turret demonstration.

EXPERIENCE

JSL Logistics, *Intern (remote)* Aug 2026 to present

- Develop staff tools and the public website, including a document-based support chatbot that retrieves source documents, photo-assisted damage review that staff verify, and a 3D cargo-loading planner that checks dimensions, rotation, stacking support and payload limits. Built with Next.js, Node.js/Express, MongoDB, Three.js and Docker through GitHub pull requests.

Mintech, Asan, South Korea, *Intern, Manufacturing Technology* Jul 20 to Aug 19, 2026

- Assisted with manufacturing OLED deposition equipment; learned the structure and process flow of sputtering thin-film deposition equipment; assisted with dimensional checks using vernier calipers and micrometers, equipment checks and workplace organization; completed 4M, 3-jeong 5S and workplace safety onboarding.

Korea Aerospace University Aerospace and Space Camp, Goyang, South Korea Aug 3 to 7, 2026

- Completed programs on flight training devices (FTD), air traffic control (ATC), CubeSat design and autonomous drone systems.

Shandong University Chinese Language and Culture Workshop, Weihai, China Jul to Aug 2025

- Practiced Mandarin daily and joined cultural activities with an international group.

LEADERSHIP AND ACTIVITIES

COSMOS Aero-Engineering Club, *Co-founder; President since Mar 2026* Sep 2025 to present

- Co-led a rocket thrust-stand project through its static-fire tests; after the first test failed to save data, corrected the recording sequence, and a later test captured a continuous thrust record reaching about 190 N and a total impulse

of about 270 N·s.
- Direct Project AEROVIEW, a suction wind tunnel designed in FreeCAD and OpenSCAD to compare NACA 0012, 2412 and 4412 airfoils; found and corrected a design-to-fabrication dimensional error by direct measurement.
- Coordinate TVC-Hopper, which combines reinforcement-learning simulation, avionics and CAD for thrust-vector control (calibration and physical validation in progress).

DAEMUN III (Daewon Model UN), *Deputy Head of Technology* 2026

- Co-developed the conference platform with Junwon Kim: public site, AI assistant, organizer dashboard, and delegate document submissions with version history, team notifications and committee-level publishing. Built the assistant's first backend, admin tools and chat widget, and expanded its FAQ to 103 entries.
- After the assistant stopped during a school-wide demonstration, called a team meeting; the team moved to a gateway-based design with a fallback provider and full-site context, and the next 51 logged queries recorded no upstream errors.

Student Council, Qingdao Daewon School, *Planning Department Vice President* 2025 to Mar 2026

- Helped plan a joint sports day for the domestic and international divisions; ran the live scoreboard at the orientation event and helped with equipment setup, rehearsals and cleanup.

Orchestrators (formerly Mechanics Club), *Vice President* Mar 2026 to present

- Student AI development team of six. Contributed to Agent Device, a palm-sized companion device for AI agents: assigned PCB design verification for the v0.3 keyring board (layout script-generated before review); wrote its order review, checked the module footprint and pin assignments against the datasheet, flagged a 1.6 mm board versus 0.8 mm case thickness mismatch, and wrote a pad-and-pin audit tool with regression tests. The order decision is pending; the board is not yet assembled and firmware has not started.
- Submitted a researcher and creator workflow for the team's Slack agent (pull request under review).
- As Mechanics Club, designed and tested a 4-bit ripple-carry adder from logic-gate ICs and LED indicators; first sketched the flight-booking concept that became Tern.

DAEMUN II (Daewon Model UN), *Delegate for Slovakia, UNCSTD* 2025

- Spoke in debate on drone weaponization and drafted clauses proposing an international drone-forensics hub.

DAEMUN I (Daewon Model UN), *Admin staff* 2024

- Supported the chairs' communication and kept committee procedure running as conference staff.

Support Group, *Group Leader (Grade 11)* 2024 to 2026

- Joined four semesters of the school's peer-mentoring program; led the group in Grade 11.

PSYCHOFISH peer-support club; Point of View club (PR team) 2024 to 2025

- Helped run an anonymous peer-support channel and designed recruitment posters; wrote rules and posters for a school-wide Manito event.

SERVICE

- School service, Nov 2024 to Jun 2026: classroom equipment management, school cleaning, orientation assistant, student council event support, and graduation ceremony support including production of the ceremony video.
- Volunteer at Qingdao Tonghang House, a welfare facility for people with intellectual disabilities, Oct to Nov 2025.

OTHER

- Airbus Aerospace Virtual Work Experience, Springpod online programme (Mar 2026); Claude Development and AI Fluency, Anthropic online course (Jul 2026).
- Participant, Goyang Mayor's Cup three-cushion billiards tournament (Jul 2026).

SKILLS

Python, TypeScript, JavaScript, Next.js, React, Node.js, Express, Tailwind CSS, Three.js, MongoDB, PostgreSQL, Docker, Git/GitHub, APIs, OpenCV, Arduino/ESP32, KiCad, FreeCAD, OpenSCAD. Languages: Korean, English, Mandarin.